the individual making the sperm, i.e., from the paternal grandparents of the child who is eventually conceived by the sperm.) The process of swapping bits of chromosome is called *crossing over*. It is very important for the whole argument of this book. It means that if you got out your microscope and looked at the chromosomes in one of your own sperms (or eggs if you are female) it would be a waste of time trying to identify chromosomes that originally came from your father and chromosomes that originally came from your mother. (This is in marked contrast to the case of ordinary body cells (see page 31).) Any one chromosome in a sperm would be a patchwork, a mosaic of maternal genes and paternal genes.

The metaphor of the page for the gene starts to break down here. In a loose-leaf binder a whole page may be inserted, removed or exchanged, but not a fraction of a page. But the gene complex is just a long string of nucleotide letters, not divided into discrete pages in an obvious way at all. To be sure, there are special symbols for END OF PROTEIN CHAIN MESSAGE and START OF PROTEIN CHAIN MESSAGE written in the same four-letter alphabet as the protein messages themselves. In between these two punctuation marks are the coded instructions for making one protein. If we wish, we can define a single gene as a sequence of nucleotide letters lying between a START and an END symbol, and coding for one protein chain. The word *cistron* has been used for a unit defined in this way, and some people use the word gene interchangeably with cistron. But crossing-over does not respect boundaries between cistrons. Splits may occur within cistrons as well as between them. It is as though the architect's plans were written out, not on discrete pages, but on 46 rolls of ticker tape. Cistrons are not of fixed length. The only way to tell where one cistron ends and the next begins would be to read the symbols on the tape, looking for END OF MESSAGE and START OF MESSAGE symbols. Crossing-over is represented by taking matching paternal and maternal tapes, and cutting and exchanging matching portions, regardless of what is written on them.

In the title of this book the word gene means not a single cistron but something more subtle. My definition will not be to everyone's taste, but there is no universally agreed definition of a gene. Even if there were, there is nothing sacred about definitions. We can define a word how we like for our own purposes, provided we do so clearly and unambiguously. The definition I want to use comes from G. C.